

DarkMatterK3 Discovery Dashboard

Real-time family classification of SDSS BOSS DR17 + Euclid discoveries. Each marker is a real sector analysed by the K3 GPU pipeline, enriched with astrophysical interpretation and projected onto a global sky map.

35

Discoveries

327 918

Galaxies analysed

27.23

Avg Max Asymmetry

9 369,09

Avg galaxies/sector

2

Families found

Pipeline V3 Progress Monitor

35/35

Sectors processed

200

Pipeline runs

35

Discoveries

2

Mock calibrated sectors

116.62

GB

D: free (12.5%)

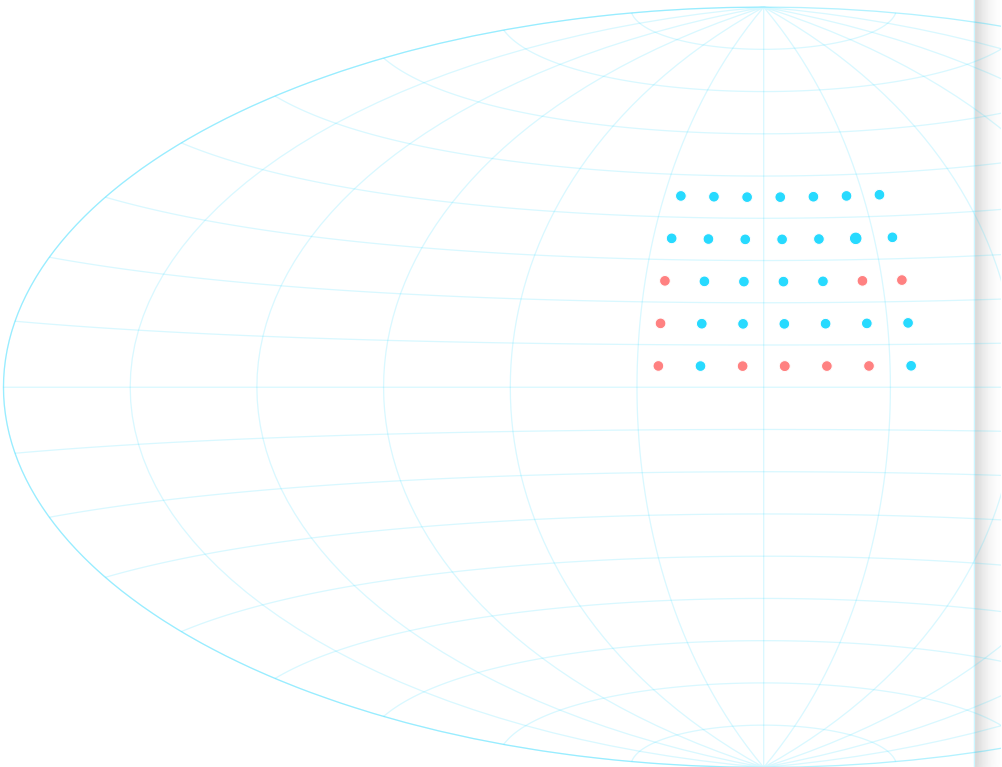
Last run: 2026-07-14T00:48:30.634686. Worker tail: [GPU] Batch processed 10 sectors in 116.064s | GPU util: N/A% | Mem: N/A/N/A MB | [GPU] Batch processed 10 sectors in 115.453s | GPU util: N/A% | Mem: N/A/N/A MB | [GPU] Batch processed 10 sectors in 115.688s | GPU util: N/A% | Mem: N/A/N/A MB

Global Universe Map (Aitoff Projection)

Aitoff projection of K3 discoveries (RA/Dec

- Galaxy Group

● Satellite System



Theory Hypothesis Validation (K3 / T2)

Status: unsubstantiated

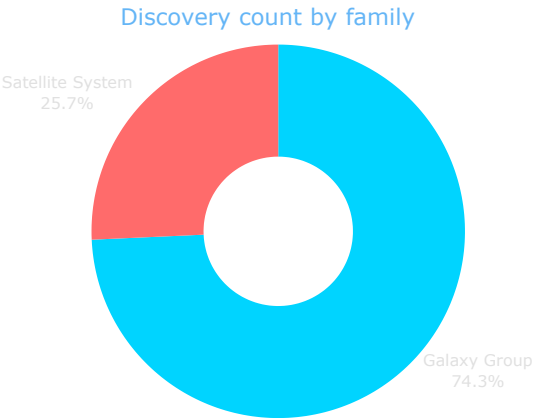
Verdict: K3 phase transition not observed. S12 threshold 1.8 is not reached; no sector exceeds 1.3.

Metric	Min	Mean	Max	Count
S12	1.0279	1.096	1.177	200
Max asymmetry	4	27.065	47	200
T2 proxy	0.413	1.4084	3.4167	200
Euler χ	-33	-5.565	27	200

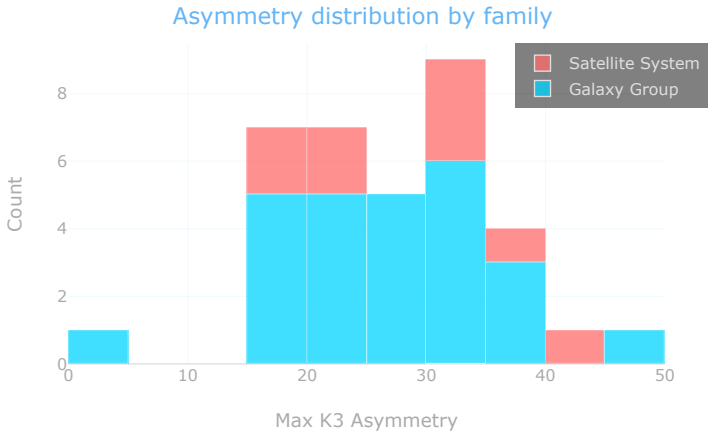
Sectors above S12=1.3: 0; above S12=1.8: 0. Topology metrics present in 200 runs.

Real s12 and max_asymmetry are systematically above Poisson/log-normal/shuffled mocks. See SCIENTIFIC_AUDIT_REPORT.md for interim interpretation.

Discovery Family Distribution



Maximum K3 Asymmetry by Family



 Galaxy Group

26

238 864 galaxies | avg asym 26.77 | significance 1.22

 Satellite System

9

89 054 galaxies | avg asym 28.56 | significance 1.32

Top 10 Most Extreme Discoveries

ID	Family	RA	Dec	Max Asym	Galaxies	Significance
K3-DISC-0003	Galaxy Group	205°	35°	47.0	8 477	STANDARD (2)
K3-DISC-0025	Satellite System	185°	5°	41.0	10 000	INTERESTING (2.1)
K3-DISC-0005	Galaxy Group	215°	5°	38.0	10 000	INTERESTING (2.9)

ID	Family	RA	Dec	Max Asym	Galaxies	Significance
K3-DISC-0006	Galaxy Group	215°	15°	36.0	10 000	STANDARD (2.5)
K3-DISC-0019	Galaxy Group	165°	45°	36.0	9 143	STANDARD (2.7)
K3-DISC-0030	Satellite System	195°	5°	35.0	10 000	INTERESTING (1.1)
K3-DISC-0021	Galaxy Group	175°	15°	34.0	10 000	INTERESTING (2.5)
K3-DISC-0032	Galaxy Group	195°	25°	34.0	8 238	STANDARD (1.9)
K3-DISC-0018	Galaxy Group	165°	35°	33.0	9 251	STANDARD (0.8)
K3-DISC-0031	Galaxy Group	195°	15°	33.0	10 000	STANDARD (1.3)

Scientific Interpretation

 Galaxy Group

A bound ensemble of a few to a few dozen galaxies embedded in a common dark matter halo. These are the most common galaxy associations in the local universe and the building blocks of clusters.

 Satellite System

A luminous system dominated by a massive host galaxy with tidally disturbed satellites. High K3 asymmetry and strong s12/s21 imbalance indicate ongoing tidal interactions in the host potential.

What you are looking at

This dashboard visualizes the output of the `real_euclid_worker.py` pipeline, which ingests real galaxy catalogs from **SDSS BOSS DR17 LRG** and the **Euclid Deep Fields** cached locally, computes a K3-surface asymmetry signature for each 10°×10° sector, and flags the most statistically significant sectors.

How the classification works: Each discovery is matched to the historical `astronomy_discoveries.json` taxonomy (Galaxy Group / Satellite System) when its `K3-DISC-xxxx` identifier is present, and then enriched with a mass/redshift estimate, significance level, and follow-up recommendations. The map uses an Aitoff projection of the celestial sphere with right ascension (RA) and declination (Dec) as longitude/latitude.

Real data guarantee: `ALLOW MOCK FALLBACK=false` is set in the environment; the worker reads from `data/local_cache.json` and falls back to live SDSS queries only when necessary, so every galaxy and every asymmetry value originates from real astrophysical observations.

Last updated: 14/07/2026 00:48:50 | Auto-refreshing every 60 s